

WAYNE WU

wayne.wuuuu@gmail.com

www.wuwayne.com

www.linkedin.com/in/wayne-wu

HIGHLIGHTS

- **Focus:** 3D visualization, simulation, procedural graphics, real-time rendering, immersive digital experiences, and pipeline engineering
- **Hobbies:** traveling, bartending, cinematography

SKILLS

- **Tools:** Houdini, Blender, Unity, Omniverse, UE5, Vulkan, WebGPU, PyTorch, AWS
- **Languages:** C++, Python, CUDA/Warp, GLSL, C#, Java, JavaScript, SQL, HTML

PROFESSIONAL EXPERIENCE

Senior Graphics Engineer (Technical Lead), Genies (Los Angeles, CA)

Sep 2023 — Dec 2025

- Architected and launched a 0→1 cloud-native auto-rigging platform for 5000+ AI-generated 3D avatars and wearables, defining system architecture across graphics libraries, asset pipeline, and ML integration.
- Modeled GPU and cloud compute usage across AWS workloads, forecasting per-asset processing cost and monthly infrastructure spend, and informing scaling strategy under production growth constraints.
- Integrated 3D software and pipeline components (Python, C++, Houdini, Blender) for geometry processing, procedural rigging, simulation, and synthetic data generation for ML training.

Lecturer (GPU Programming), University of Pennsylvania (Philadelphia, PA)

Sep 2023 — Dec 2023

- Designed and taught graduate-level coursework for 30 students on advanced GPU algorithms, frameworks, and architecture, covering the latest features in CUDA, Vulkan, and WebGPU.

R&D Engineer (Production), Industrial Light & Magic (San Francisco, CA)

Jun 2021 — Sep 2021

- Presented Blue Sky Studios' procedural OpenUSD workflows at SIGGRAPH 2021 and adapted these concepts to ILM's pipeline, shaping the studio's long-term USD strategy across departments.

Technical Director (Production Technology), Blue Sky Studios (Greenwich, CT)

Aug 2019 — Jan 2021

- Architected Houdini-based pipeline systems (Python) with PDG integration and simplified pipeline bridges, enabling technical artists to create parallelized setups across 30,000+ show assets.
- Established workflows in OpenUSD and Solaris (VEX, Python) for rigid-body, volumetric, and procedural effects, including data translation, scene organization, caching, shading, and rendering with RenderMan.

3D Software Developer (Co-op), SideFX (Toronto, ON)

Fall 2017 & Summer 2018

- Developed the material-based fracturing toolkit in Houdini with newly designed rigid-body dynamics workflows based on client requirements.

ACADEMIC & PERSONAL PROJECTS

Metadrobe (Digital Fashion), Spring 2023

- Designed and prototyped a digital fashion and virtual try-on system using advanced 3D tools (3D Scanning, Clo3D, Houdini, UE5, Omniverse) informed by 70+ expert interviews.

Wiggly (Authoring Tool), Spring 2022

- Implemented an optimization-based algorithm for animating deformable objects in Houdini (C++), and designed a modular and procedural workflow (Python, VEX) for artists to easily create animations.

Real-time Crowd Simulation, Fall 2021

- Developed the first WebGPU-based interactive crowd simulation using position-based dynamics, compute shaders (WGSL, JS) and optimized render pipelines to support 10,000+ agents in real time.

EDUCATION

University of Pennsylvania, Philadelphia, PA, USA

Master of Science in Engineering, **Computer Graphics and Game Technology**, University of Pennsylvania, 2023

- Teaching Assistant for GPU Programming and Architecture, Fall 2022

University of Waterloo, Waterloo, ON, Canada

Bachelor of Applied Science (Honors), **Systems Design Engineering**, 2019

- Graduated with Distinction, 2019 | Engineering Faculty/Staff Upper Year Scholarship, 2018